MOBILE TRACKING SYSTEM

A PROJECT REPORT

Submitted by

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in partial fulfillment of requirements for the award of the course

CGB1201 - JAVA PROGRAMMING

In

INFORMATION TECHNOLOGY

K. RAMAKRISHNAN COLLEGE OF TECHNOLOGY

(An Autonomous Institution, affiliated to Anna University Chennai and Approved by AICTE, New Delhi)

SAMAYAPURAM – 621 112

NOVEMBER- 2024

K. RAMAKRISHNAN COLLEGE OF TECHNOLOGY (AUTONOMOUS)

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BONAFIDE CERTIFICATE

Certified that this project report on **" MOBILE TRACKING SYSTEM "** is the bonafide work of **SANJEEV KUMAR R (2303811720521044)** who carried out the project work during the academic year 2024 - 2025 under my supervision.

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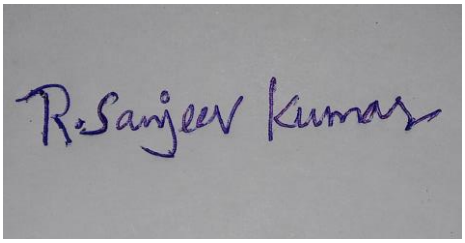
I declare that the project report on **“MOBILE TRACKING SYSTEM”** is the result of original work done by us and best of our knowledge, similar work has not been submitted to **“ANNA UNIVERSITY CHENNAI”** for the requirement of Degree of **BACHELOR OF TECHNOLOGY**. This project report is submitted on the partial fulfilment of the requirement of the completion of the course **CGB1201 - JAVA PROGRAMMING.**

Signature



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Place: Samayapuram

Date:04.12.2024

ACKNOWLEDGEMENT

It is with great pride that I express our gratitude and in-debt to our institution "**K.Ramakrishnan College of Technology (Autonomous)**", for providing us with the opportunity to do this project.

I glad to credit honourable chairman **Dr. K. RAMAKRISHNAN**, **B.E.,** for having provided for the facilities during the course of our study in college.

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I render our sincere thanks to Course Coordinator and other staff members for providing valuable information during the course.

I wish to express our special thanks to the officials and Lab Technicians of our departments who rendered their help during the period of the work progress.

VISION OF THE INSTITUTION

To serve the society by offering top-notch technical education on par with global standards

MISSION OF THE INSTITUTION

- Be a center of excellence for technical education in emerging technologies by exceeding the needs of the industry and society.
- Be an institute with world class research facilities
- Be an institute nurturing talent and enhancing the competency of students to transform them as all-round personality respecting moral and ethical values

VISION OF DEPARTMENT

To be a center of eminence in creating competent software professionals with research and innovative skills.

MISSION OF DEPARTMENT

M1: Industry Specific: To nurture students in working with various hardware and software platforms inclined with the best practices of industry.

M2: Research: To prepare students for research-oriented activities.

M3: Society: To empower students with the required skills to solve complex technological problems of society.

PROGRAM EDUCATIONAL OBJECTIVES

1. PEO1: Domain Knowledge

To produce graduates who have strong foundation of knowledge and skills in the field of Computer Science and Engineering.

2. PEO2: Employability Skills and Research

To produce graduates who are employable in industries/public sector/research organizations or work as an entrepreneur.

3. PEO3: Ethics and Values

To develop leadership skills and ethically collaborate with society to tackle real-world challenges.

PROGRAM SPECIFIC OUTCOMES (PSOs)

PSO 1: Domain Knowledge

To analyze, design and develop computing solutions by applying foundational concepts of Computer Science and Engineering.

PSO 2: Quality Software

To apply software engineering principles and practices for developing quality software for scientific and business applications.

PSO 3: Innovation Ideas

To adapt to emerging Information and Communication Technologies (ICT) to innovate ideas and solutions to existing/novel problems

PROGRAM OUTCOMES (POs)

Engineering students will be able to:

1. **Engineering knowledge:** Apply the knowledge of mathematics, science, engineering fundamentals, and an engineering specialization to the solution of complex engineering problems.
2. **Problem analysis:** Identify, formulate, review research literature, and analyze complex engineering problems reaching substantiated conclusions using first principles of mathematics, natural sciences, and engineering sciences
3. **Design/development of solutions:** Design solutions for complex engineering problems and design system components or processes that meet the specified needs with appropriate consideration for the public health and safety, and the cultural, societal, and environmental considerations
4. **Conduct investigations of complex problems:** Use research-based knowledge and research methods including design of experiments, analysis and interpretation of data, and synthesis of the information to provide valid conclusions

5. **Modern tool usage:** Create, select, and apply appropriate techniques, resources, and modern engineering and IT tools including prediction and modeling to complex engineering activities with an understanding of the limitations
6. **The engineer and society:** Apply reasoning informed by the contextual knowledge to assess societal, health, safety, legal and cultural issues and the consequent responsibilities relevant to the professional engineering practice
7. **Environment and sustainability:** Understand the impact of the professional engineering solutions in societal and environmental contexts, and demonstrate the knowledge of, and need for sustainable development
8. **Ethics:** Apply ethical principles and commit to professional ethics and responsibilities and norms of the engineering practice.
9. **Individual and team work:** Function effectively as an individual, and as a member or leader in diverse teams, and in multidisciplinary settings.
10. **Communication:** Communicate effectively on complex engineering activities with the engineering community and with society at large, such as, being able to comprehend and write effective reports and design documentation, make effective presentations, and give and receive clear instructions.
11. **Project management and finance:** Demonstrate knowledge and understanding of the engineering and management principles and apply these to one's own work, as a member and leader in a team, to manage projects and in multidisciplinary environments.
12. **Life-long learning:** Recognize the need for, and have the preparation and ability to engage in independent and life-long learning in the broadest context of technological change.

ABSTRACT

A mobile tracking system is a technology designed to monitor, manage, and update the location of mobile devices using straightforward functionalities. It allows users to uniquely identify devices through their International Mobile Equipment Identity (IMEI) numbers and retrieve their stored location information. The system uses basic features like adding devices, tracking their current location, and manually updating the location when required. With a simple interface and easy-to-use input methods, this system is highly user-friendly, making it accessible for individuals and organizations alike. The tracking mechanism relies on secure data handling and efficient management of device information. Although it operates on a basic level, such systems form the groundwork for building more advanced mobile tracking solutions. These can be expanded to include real-time GPS tracking, integration with mapping services, and robust data storage in a database for persistent access.

ABSTRACT WITH POs AND PSOs MAPPING

CO 5 : BUILD JAVA APPLICATIONS FOR SOLVING REAL-TIME PROBLEMS.

ABSTRACT	POs MAPPED	PSOs MAPPED
A mobile tracking system is a technology designed to monitor, manage, and update the location of mobile devices using straightforward functionalities. It allows users to uniquely identify devices through their International Mobile Equipment Identity (IMEI) numbers and retrieve their stored location information. The system uses basic features like adding devices, tracking their current location, and manually updating the location when required. With a simple interface and easy-to-use input methods, this system is highly user-friendly, making it accessible for individuals and organizations alike. The tracking mechanism relies on secure data handling and efficient management of device information. Although it operates on a basic level, such systems form the groundwork for building more advanced mobile tracking solutions. These can be expanded to include real-time GPS tracking, integration with mapping services, and robust data storage in a database for persistent access.	**PO1 -3** **PO2 -3** **PO3 -3** **PO4 -3** **PO5 -3** **PO6 -3** **PO7 -3** **PO8 -3** **PO9 -3** **PO10 -3** **PO11-3** **PO12 -3**	**PSO1 -3** **PSO2 -3** **PSO3 -3**

Note: 1- Low, 2-Medium, 3- High

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CHAPTER 1
INTRODUCTION

1.1 Objective

The objective of this project is to simulate a Mobile Tracking System that allows users to track mobile locations, report lost phones, and manage stored data. Users can input a valid 10-digit mobile number to either simulate tracking its geographical location using random latitude and longitude or register it as lost. The system maintains a list of tracked and lost mobile numbers, which can be displayed on demand. A user-friendly graphical interface ensures smooth interactions, while features like input validation and confirmation dialogs improve usability. This program demonstrates key concepts in Java, including GUI development, event handling, and data storage using collections.

1.2 Overviews

This project in Java-based on Mobile Tracking System built using the AWT and Swing libraries for a graphical user interface. It allows users to perform three main tasks: track a mobile's location, report a lost mobile, and display stored data related to these actions. Mobile locations are simulated with random latitude and longitude values, and both tracked and lost mobile numbers are stored in dynamic lists using the ArrayList class. The application ensures input validation for 10-digit mobile numbers and provides clear outputs in a text area. A simple yet efficient GUI design includes buttons and input fields, along with event handling for user actions. Additionally, the program incorporates confirmation dialogs to enhance user interaction and ensure data safety when exiting. It serves as an educational example of integrating multiple Java concepts like GUI development, event handling, and collections.

1.3 Java Programming Concepts

The Mobile Tracking System leverages several core Java concepts to achieve its functionality. It utilizes Object-Oriented Programming (OOP) principles, such as encapsulation, where the logic is contained within a class, and inheritance, as the main class extends Frame to inherit GUI-related properties and methods. For GUI Development, AWT components like Frame, Label, TextField, TextArea, and Button are employed, along with layout managers like FlowLayout, while Swing's JOptionPane is used for confirmation dialogs. The project incorporates Event Handling through ActionListener to manage button clicks and WindowAdapter for handling the window closing event. The Collection Framework is applied using ArrayList to dynamically store tracked and lost mobile numbers. Simulated mobile location tracking is implemented using the Random class to generate random latitude and longitude values. Input Validation ensures the mobile number is valid using String.matches(), while String Manipulation methods such as trim() and append() are used to process user input and format output. These combined concepts ensure the system is robust, interactive, and user-friendly.

Mobile Tracking System related Concepts:

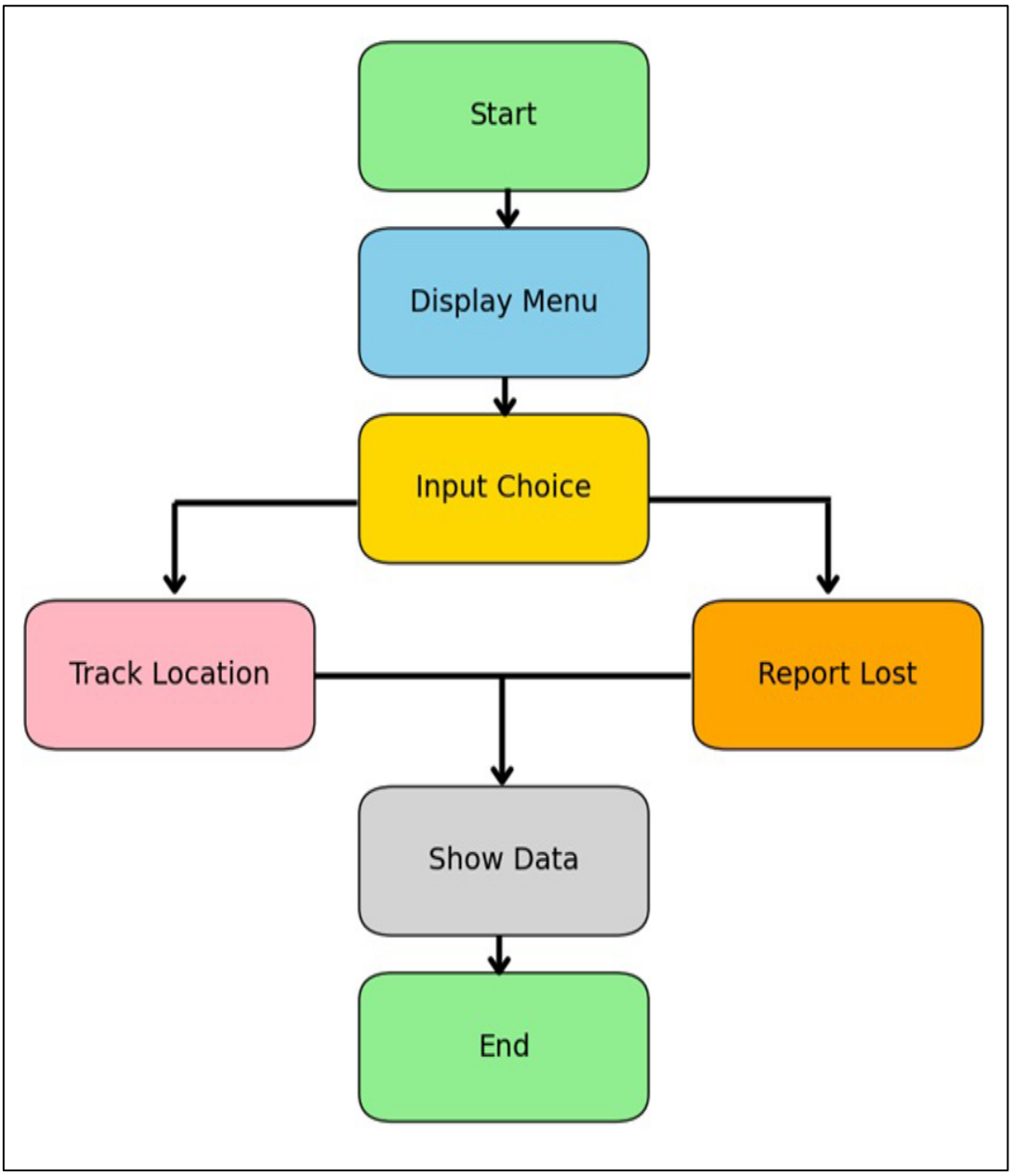
The Mobile Tracking System effectively combines multiple functionalities to provide a seamless user experience. It features a Mobile Tracking Simulation, where random geographical coordinates are generated for valid mobile numbers to mimic tracking. The Lost Mobile Reporting functionality enables users to report lost phones, with the data stored for future reference. The system employs efficient Data Management by maintaining two lists to organize and retrieve tracked and lost mobile numbers. A User-Friendly Interface is developed using graphical components, allowing users to easily input mobile numbers, trigger actions, and view outputs. Additionally, Confirmation and Error Handling mechanisms, such as exit confirmation dialogs and alerts for invalid inputs, ensure robustness and enhance the overall user experience.

CHAPTER 2
PROJECT METHODOLOGY

2.1 Proposed Work

The proposed work for the Mobile Tracking System project involves designing and implementing a Java-based application that simulates essential mobile tracking operations. The system will provide three primary functionalities: tracking mobile locations, reporting lost phones, and managing stored data. The work begins with System Design, where a graphical user interface (GUI) is developed using AWT components to enable user interaction. The layout will include input fields, buttons, and text areas to facilitate tracking, reporting, and data display tasks. For Mobile Tracking Simulation, random geographical coordinates will be generated to simulate the tracking of valid mobile numbers. The Lost Mobile Reporting functionality will allow users to report lost phones, with details securely stored in a dynamic data structure. Data Management will be implemented using the ArrayList class to manage two categories of mobile numbers: tracked and lost, with the option to display this data in a structured format. Validation and Error Handling will ensure that only valid 10-digit mobile numbers are accepted, providing users with meaningful feedback for invalid inputs. User Interaction Enhancements will include confirmation dialogs for actions like exiting the program, along with real-time updates to the text area for displaying results and stored data. The project will undergo Final Testing and Debugging to ensure the system is functional, user-friendly, and free from major bugs, optimizing the interface and logic for performance and usability. This work aims to showcase core Java programming concepts such as GUI development, event handling, and data storage, offering an interactive experience for the end user.

2.2 Block Diagram



CHAPTER 3
MODULE DESCRIPTION

3.1 Start Module

This module initializes the program and sets up the necessary components such as the GUI and backend structures (e.g., ArrayList for data storage). prepares the system to handle user input and interactions.

3.2 Display Menu Module

This module shows the main menu with available options such as tracking location, reporting a lost mobile, and displaying stored data. The user can select their desired action through this menu.

3.3 Input Choice Module

This module captures the user's choice from the menu. It processes the input and directs the flow of the program to the appropriate functionality, such as tracking, reporting, or displaying data.

3.4 Functional (Track Location and Report Lost) Module

This sub-module simulates mobile tracking by generating random latitude and longitude values for a valid 10-digit mobile number and stores it in the tracking data.

3.5 Show Data and End Module

This module retrieves and displays all stored data (tracked mobile numbers and reported lost numbers) in an organized format.

CHAPTER 4

4.1 CONCLUSION

The Mobile Tracking System project successfully demonstrates the integration of core Java concepts such as GUI development, event handling, and data management using ArrayList. By providing functionalities like mobile location tracking, lost mobile reporting, and stored data display, the program offers a user-friendly simulation of a basic mobile tracking system. Input validation and confirmation dialogs enhance the reliability and usability of the application. This project highlights the practical application of object-oriented programming principles and Java's graphical capabilities to solve real-world problems. Overall, it serves as an excellent educational tool for understanding how to build interactive and functional Java-based applications.

4.2 FUTURE AND SCOPE

The future scope of the Mobile Tracking System includes expanding its capabilities to offer more advanced features and improving its overall functionality. One potential enhancement is the integration of real-time GPS tracking, allowing the system to provide actual location updates for mobile devices, rather than relying on simulated data. Additionally, the application could be integrated with external databases or cloud services for persistent storage of tracked and lost mobile data, ensuring that users can access their reports even after system restarts. Implementing multi-device support and synchronizing data across platforms could further enhance the system's accessibility.

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APPENDIX A

(Project Source Code)

```
import java.awt.*;
import java.awt.event.*;
import java.util.ArrayList;
import java.util.List;
import java.util.Random;
import javax.swing.JOptionPane;

public class Sanjeev extends Frame implements ActionListener {
    private List<String> trackedMobiles = new ArrayList<>();
    private List<String> lostMobiles = new ArrayList<>();
    private Random random = new Random();

    private Label label;
    private TextField inputField;
    private TextArea outputArea;
    private Button trackButton, reportLostButton, displayButton, exitButton;

    public Sanjeev() {
        // Frame setup
        setTitle("Mobile Tracking System");
        setSize(500, 400);
        setLayout(new FlowLayout());
        setResizable(false);

        // UI Components
        label = new Label("Enter Mobile Number:");
        inputField = new TextField(20);
        outputArea = new TextArea(15, 40);
        trackButton = new Button("Track Mobile Location");
        reportLostButton = new Button("Report Lost Mobile");
```

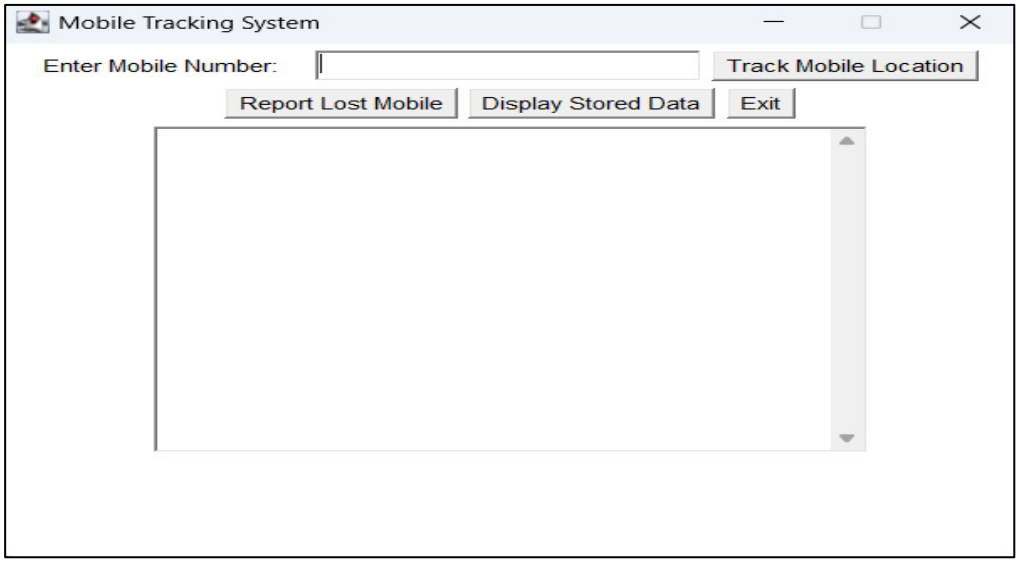
```
        displayButton = new Button("Display Stored Data");
        exitButton = new Button("Exit");

        // Adding components to the frame
        add(label);
        add(inputField);
        add(trackButton);
        add(reportLostButton);
        add(displayButton);
        add(exitButton);
        add(outputArea);

        // Event Listeners
        trackButton.addActionListener(this);
        reportLostButton.addActionListener(this);
        displayButton.addActionListener(this);
        exitButton.addActionListener(this);

        // Frame close behavior with confirmation
        addWindowListener(new WindowAdapter() {
            public void windowClosing(WindowEvent we) {
                int response = JOptionPane.showConfirmDialog(null, "Are you sure you
want to exit?", "Confirm Exit",
                        JOptionPane.YES_NO_OPTION,
JOptionPane.QUESTION_MESSAGE);
                if (response == JOptionPane.YES_OPTION) {
                    dispose();
                }
            }
        });

        setVisible(true);
    }
```

```
@Override
public void actionPerformed(ActionEvent e) {
    String action = e.getActionCommand();
    String mobileNumber = inputField.getText().trim();

    if (action.equals("Track Mobile Location")) {
        if (mobileNumber.isEmpty() || !mobileNumber.matches("\\d{10}")) {
            outputArea.setText("Please enter a valid 10-digit mobile number to track.");
            return;
        }
        double latitude = 10 + (random.nextDouble() * (20 - 10)); // Random latitude
        double longitude = 70 + (random.nextDouble() * (80 - 70)); // Random
longitude
        trackedMobiles.add(mobileNumber);
        outputArea.setText("Tracking mobile number: " + mobileNumber + "\n");
        outputArea.append(String.format("Location: Latitude %.4f, Longitude
%.4f\n", latitude, longitude));
    } else if (action.equals("Report Lost Mobile")) {
        if (mobileNumber.isEmpty() || !mobileNumber.matches("\\d{10}")) {
            outputArea.setText("Please enter a valid 10-digit mobile number to report as
lost.");
            return;
        }
        lostMobiles.add(mobileNumber);
        outputArea.setText("Lost mobile report for number " + mobileNumber + " has
been registered.\n");
        outputArea.append("Please contact your service provider for further support.");
    } else if (action.equals("Display Stored Data")) {
        outputArea.setText("==== Stored Data ====\n");
        outputArea.append("Tracked Mobile Numbers:\n");
        if (trackedMobiles.isEmpty()) {
            outputArea.append("No mobile numbers have been tracked yet.\n");
        } else {
            trackedMobiles.forEach(num -> outputArea.append(num + "\n"));
```

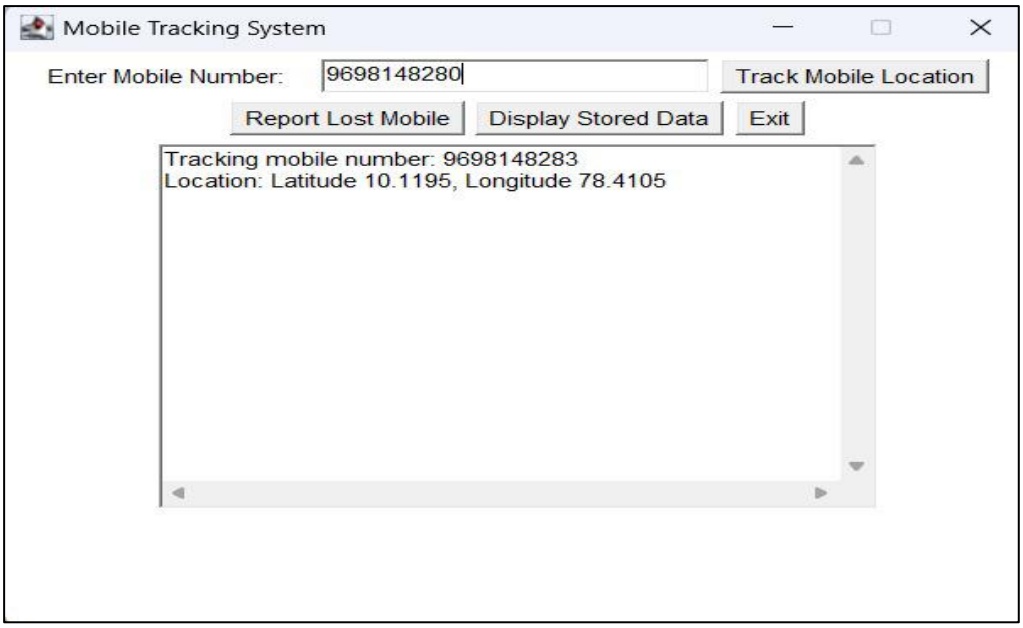
```
                }
                outputArea.append("\nLost Mobile Reports:\n");
                if (lostMobiles.isEmpty()) {
                    outputArea.append("No lost mobile reports have been filed yet.\n");
                } else {
                    lostMobiles.forEach(num -> outputArea.append(num + "\n"));
                }
            } else if (action.equals("Exit")) {
                int response = JOptionPane.showConfirmDialog(null, "Are you sure you want
to exit?", "Confirm Exit",
                        JOptionPane.YES_NO_OPTION,
JOptionPane.QUESTION_MESSAGE);
                if (response == JOptionPane.YES_OPTION) {
                    dispose();
                }
            }
        }

        public static void main(String[] args) {
            new Sanjeev();
        }
    }
```

APPENDIX B
(ScreenShort)

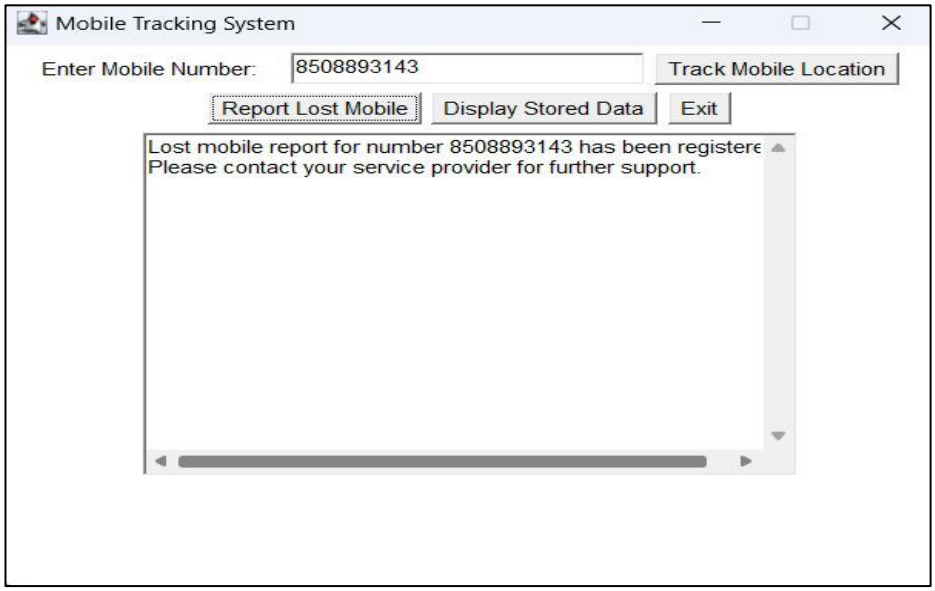
Start



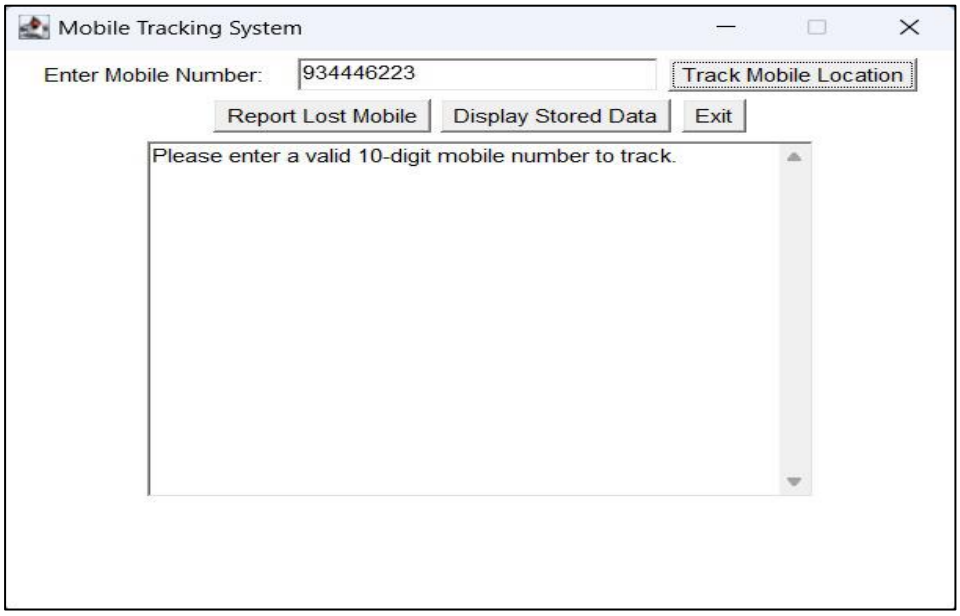
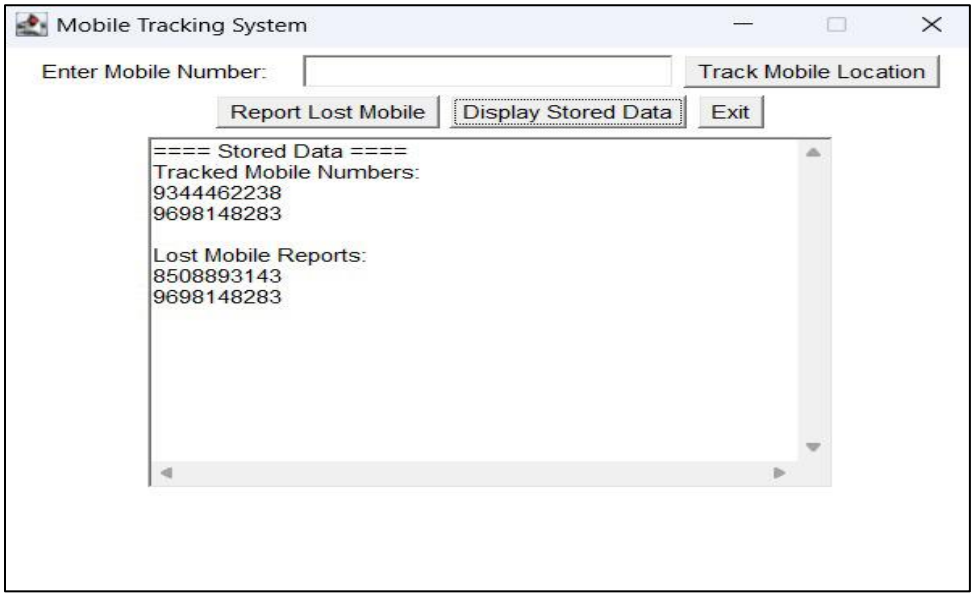
Tracking Mobile Location



Report Lost Mobile



Enter A Valid Input



Display Stored Data